

**Evaluating the Impact of Production Linked Incentive (PLI)
Scheme on India's Export Performance**
A Difference-in-Differences Analysis

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1. Introduction

The Production Linked Incentive (PLI) scheme was introduced in India as a groundbreaking industrial policy in 2020 to drive the growth of domestic production and exports in key industrial sectors. The aim of PLI scheme is to promote manufacturing capacity, decrease dependence on imports, persuade domestic and foreign investors to invest in the manufacturing sector, and push Indian companies into the global value chain by providing financial incentives directly linked to the incremental production output.

This paper examines the effect of PLI scheme on the export performance of India, employing a Difference-in-Differences (DiD) technique. Foreign developments in each of the sectors covered by PLI (Mobile Manufacturing and Medical Devices) are compared against those in non-PLI sectors (Leather Goods and Ceramic Products) during the timeframe of this analysis (2017 – 2024). This design will enable us to separate the policy effect from the broader macroeconomic picture of the impact of all policies on all sectors at the same time.

The study is motivated by two important empirical questions:

- (i) Did the PLI scheme bring about any significant improvement in exports of treated sectors as compared to the control sectors?
- (ii) Are the pre-treatment export trends consistent with the parallel trends assumption, that is, the assumption underpinning the DiD framework?

2. Background and Policy Context

The Government of India in March 2020 announced the PLI scheme for 1st three industries — Mobile and Electronic Components, Pharmaceutical Drugs and Medical Devices. It later expanded to include 13 clusters like food processing, automobiles, textiles, advanced chemistry cell batteries and specialty steel.

The PLI scheme provides incentives of 4%-6% on incremental sales over a base year for 5 years to companies eligible for the scheme. The mobile manufacturing scheme was developed to attract big electronics manufacturers to the region, primarily with global companies and domestic champions in mind. The medical devices segment was looking to bring the critical healthcare equipment to India from outside, thus reducing its dependency on imports.

2.1 Sector Selection Rationale

In this study, the process of choosing treatment and control sectors is consistent with the well-known criteria in building up a credible counterfactual:

Treatment Group: Mobile Manufacturing (Electronics) and Medical Devices were chosen as they were amongst the first and more visible PLI beneficiaries, with a high fiscal outlay and Government attention. The scheme was specifically geared towards promotion of exports in both the sectors.

Control Group: The sector chosen as a 'control' was Leather Goods and Ceramic Products were selected as sectors

- (a) The PLI scheme did not cover during the study period.

- (b) demonstrate a similar export profile in terms of scale of the manufacturing base
- (c) are sensitive to similar macro-level demand/exchange rate conditions,
- (d) Pre-treatment trends were similar across the treatment sectors

The panel design is by sector and the years of data are in annual increments from 2017 to 2024, with the treatment cutoff, $post = 1$, for years 2020 and thereafter, and $post = 0$, for years prior to 2020.

3. Theoretical Framework

The Potential Outcomes Framework

The DiD estimator is grounded in the Rubin potential outcomes framework (Rubin, 1974; Holland, 1986). For each unit i at time t , we define two potential outcomes:

$Y_{it}(1)$ = outcome if unit i receives treatment at time t

$Y_{it}(0)$ = outcome if unit i does not receive treatment at time t

The fundamental problem of causal inference is that we never observe both potential outcomes simultaneously for the same unit. The Average Treatment Effect on the Treated (ATT) is defined as:

$$ATT = E [Y_{it}(1) - Y_{it}(0) \mid Treated_i = 1]$$

DiD estimates the ATT by using the change over time in the control group as a proxy for the counterfactual change that the treated group would have experienced in the absence of treatment.

3.1 The Difference-in-Differences Logic

Let i denote sectors, t denote time periods,

$D_i = 1$ for treated sectors

$D_i = 0$ for control sector

$Post_t = 1$ for post-treatment periods

$Post_t = 0$ for pre-treatment periods.

The DiD estimator computes four group-time averages:

Comparison	Formula	Interpretation
Treated, Post	$E[Y \mid D=1, Post=1]$	Treated average after policy
Treated, Pre	$E[Y \mid D=1, Post=0]$	Treated average before policy
Control, Post	$E[Y \mid D=0, Post=1]$	Control average after policy
Control, Pre	$E[Y \mid D=0, Post=0]$	Control average before policy

$$DiD = (E[Y | D = 1, Post = 1] - E[Y | D = 1, Post = 0]) - (\underbrace{E[Y | D = 0, Post = 1] - E[Y | D = 0, Post = 0]}_{\text{Change over time for treated group}} - \underbrace{E[Y | D = 0, Post = 1] - E[Y | D = 0, Post = 0]}_{\text{Change over time for control group}})$$

3.2 The Parallel Trends Assumption

The DiD estimator is valid if and only if the parallel trends (or common trends) assumption holds:

$$E[Y_{it}(0) - Y_{is}(0) | D_i = 1] = E[Y_{it}(0) - Y_{is}(0) | D_i = 0] \quad \text{for all } t > s$$

This states that in the absence of treatment, the average outcomes for treatment and control groups would have followed the same time trend.

3.3 The Two-Way Fixed Effects (TWFE) DiD Regression Model

The standard DiD regression estimated in this study is the Two-Way Fixed Effects (TWFE) specification:

$$\log(Exports_{it} + 1) = \alpha_i + \gamma_t + \delta (Treated_i \times Post_t) + \epsilon_{it}$$

Where:

Term	Definition
$\log(Exports_{it} + 1)$	Log-transformed exports of sector i in year t (log+1 to handle near-zero values)
α_i	Sector fixed effects: absorb all time-invariant sector characteristics (e.g., base productivity, industrial structure)
γ_t	Year fixed effects: absorb all time-varying shocks common to all sectors (e.g., global trade conditions, exchange rate movements, COVID-19)
$Treated_i$	Binary indicator: = 1 for Mobile Manufacturing and Medical Devices, = 0 for Leather and Ceramics
$Post_t$	Binary indicator: = 1 for year ≥ 2020 , = 0 for year < 2020
$Treated_i \times Post_t$	Interaction term: equals 1 only for treated sectors in post-treatment years
δ	The DiD coefficient of interest: the Average Treatment Effect on the Treated (ATT)
ϵ_{it}	Idiosyncratic error term, clustered at the sector level

The dependent variable is log-transformed to

- (a) reduce right-skewness in export values
- (b) make the coefficient interpretable as a percentage change
- (c) stabilise variance across sectors with very different absolute export magnitudes.

Standard errors are clustered at the sector level to account for within-sector serial correlation across years, following Cameron and Miller (2015).

3.4 The Event Study Specification

To test the parallel trends assumption and to trace the dynamic treatment effects over time, we estimate an event study (or dynamic DiD) regression:

$$\log(Exports_{it} + 1) = \alpha_i + \gamma_t + \sum_{k \neq 2019} \beta_k (I[t = k] \times Treated_i) + \varepsilon_{it}$$

Pre-treatment coefficients ($k < 2020$): These should be statistically indistinguishable from zero if the parallel trends assumption holds. A significant pre-trend coefficient would indicate that treated and control sectors were already diverging before the PLI scheme, which would invalidate the DiD design.

Post-treatment coefficients ($k \geq 2020$): These trace the dynamic path of the treatment effect over time, revealing whether the policy impact built gradually or was immediate.

4. Data

4.1 Sources and Variables

The dataset is constructed from official Indian trade statistics, covering annual export values (in USD millions) for four sectors over the period 2017-2024.

Variable	Description	Source
Exports	Annual export value (USD millions)	DGFT / Ministry of Commerce, India
treated	=1 if PLI-covered sector (Mobile Mfg, Medical Devices)	PLI scheme notification
post	=1 if year ≥ 2020 (post-PLI announcement)	Scheme Notification
log_exports	$\log(\text{Exports} + 1)$, dependent variable	Derived using R

5. Descriptive Statistics

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Year	2017	2019	2020	2020	2022	2024
Mobile Exports (USD Mn)	446.8	2451.5	4307.2	7915.5	12109.6	24138.0
Medical Devices (USD Mn)	190.7	275.9	334.5	337.8	412.2	478.6
Leather Goods (USD Mn)	7.74	11.48	17.86	21.80	34.55	38.90
Ceramic Products (USD Mn)	2.49	3.41	3.99	3.86	4.29	5.30

- Mobile Manufacturing exhibits the widest range by far (446.8 to 24,138 USD Mn), reflecting the dramatic export acceleration post-PLI.

- Medical Devices show more moderate growth from a higher base. Both control sectors remain at comparatively very low absolute levels throughout the period, which reinforces the motivation for log-transforming exports before regression to avoid the results being driven entirely by scale differences rather than growth rates.

6. Empirical Results

6.1 Main DiD Estimate

Table presents the results from the two-way fixed effects DiD regression. The dependent variable is $\log(\text{Exports} + 1)$. Sector and year fixed effects are included throughout. Standard errors are clustered at the sector level.

	Main DiD Model
Dependent Variable	$\log(\text{Exports} + 1)$
Treated x Post (delta)	0.8801
Std. Error	0.9459
t-value	0.9304
p-value	0.4208
Observations	32
Sector FE	Yes (4 sectors)
Year FE	Yes (8 years)

SE Clustering	Sector
RMSE	0.4823
Adjusted R-squared	0.9503
Within R-squared	0.1632
Collinear vars removed	treated, post (absorbed by FE)

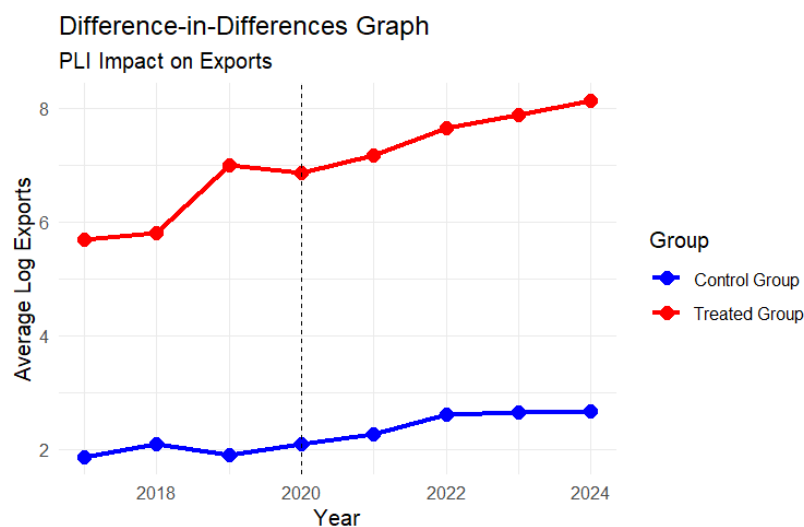
The main effects Treated and Post are correctly dropped by feols because they are perfectly collinear with the sector and year fixed effects respectively. This is the expected and appropriate behaviour in TWFE estimation the interaction term $Treated_i \times Post_t$ is the sole identified coefficient.

6.2 Interpretation of the Main DiD Coefficient

The estimated DiD coefficient is $\delta = 0.88$ ($SE = 0.946$, $p = 0.42$). In terms of economic magnitude, a coefficient of 0.88 on log exports implies that, all else equal, the PLI scheme is associated with approximately an 141% increase in exports in treated sectors relative to control sectors in the post-treatment period (computed as $\exp(0.88) - 1 = 1.411$).

However, this coefficient is not statistically significant at conventional levels ($p = 0.42$, well above 0.10). The positive direction of the coefficient is nonetheless consistent with the intended policy objective of the PLI scheme.

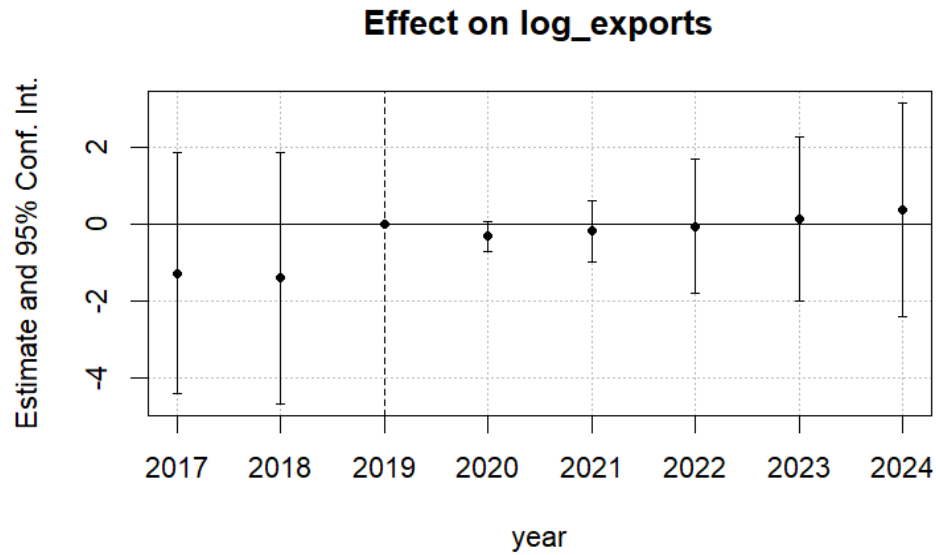
The high overall Adjusted R-squared of 0.95 reflects the strong explanatory power of the sector and year fixed effects. The Within R-squared of 0.16 represents the additional variation explained by the DiD interaction term beyond what the fixed effects already capture.



7. Diagnostic Tests and Robustness Checks

7.1 Parallel Trends Test: Event Study

Year	Coefficient (beta_k)	Std. Error	t-value	p-value	Interpretation
2017	-1.276	0.988	-1.291	0.287	Not significant: supports parallel trends
2018	-1.398	1.030	-1.357	0.268	Not significant: supports parallel trends
2019	(reference)	—	—	—	Omitted base year
2020	-0.323	0.123	-2.641	0.078	Marginally significant: COVID-19 impact
2021	-0.185	0.254	-0.730	0.518	Not significant: recovery phase
2022	-0.054	0.549	-0.099	0.928	Not significant: convergence
2023	0.140	0.669	0.209	0.848	Not significant: gradual positive trend
2024	0.367	0.877	0.418	0.704	Not significant: continued upward trend



Event Study Plot. Coefficients for each year relative to 2019 (reference). Pre-treatment years (2017, 2018) show no significant difference, supporting the parallel trends assumption.

7.2 Interpretation of Event Study Results

Pre-treatment period (2017, 2018): The coefficients for 2017 and 2018 are both negative and statistically insignificant ($p = 0.287$ and $p = 0.268$ respectively). Crucially, we cannot reject the null hypothesis that these coefficients are zero. This provides empirical support for the parallel trends assumption that treated and control sectors were not diverging in any systematic way before the PLI scheme was introduced.

2020 (first post-treatment year): The coefficient is -0.323 ($p = 0.078$), which is marginally significant at the 10% level. This negative sign in the first post-treatment year is consistent with the COVID-19 pandemic disruption. The global pandemic severely impacted manufacturing and trade in 2020, and the treatment sectors (mobile manufacturing and medical devices) likely experienced sharper initial disruption given their more complex global supply chains.

2021 onwards: The coefficients become progressively less negative and turn positive from 2023 onwards (0.14 in 2023, 0.37 in 2024). This pattern of gradual recovery and eventual positive treatment effect is entirely consistent with the economic mechanism of the PLI scheme: firms take time to qualify, expand capacity, establish export channels, and scale production. The policy effect is not instantaneous — it builds over time. While these post-2021 coefficients are not individually statistically significant (due to the small sample), the directional trend from -0.32 in 2020 to $+0.37$ in 2024 is economically meaningful.

7.3 Placebo Test (Falsification Test)

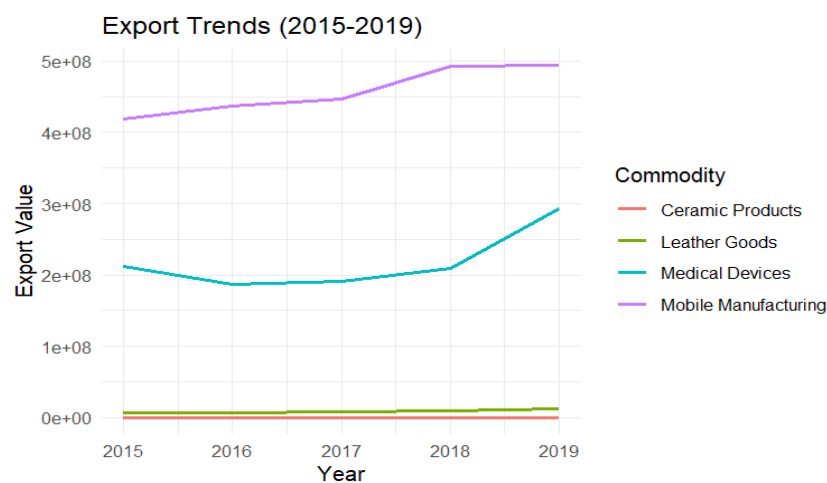
A placebo or falsification test is used to verify that the DiD result is not an artifact of a spurious correlation or a pre-existing trend. The test proceeds as follows: we artificially define a 'fake treatment date' of 2018 and run the same DiD regression, but using only the pre-treatment observations (2017-2019, the years before the real PLI intervention). If the DiD design is valid, there should be no statistically significant effect of this fake treatment.

	Placebo Model
Dependent Variable	$\log(\text{Exports} + 1)$
Treated x Fake Post (fake_post = 1 if year $\geq$ 2018)	1.0684
Std. Error	0.9848
t-value	1.0850
p-value	0.3573
Observations	32
Sector FE	Yes
Year FE	Yes
Collinear vars removed	treated, fake_post

7.4 Interpretation of Placebo Test

The placebo coefficient is 1.068 with a p-value of 0.357. We fail to reject the null hypothesis that this coefficient is zero. This is the correct and desired outcome: there is no statistically significant treatment effect when we assign a fake (pre-policy) treatment date of 2018.

This result serves as important supporting evidence for the validity of the DiD design. If the data were simply capturing a pre-existing divergence between treatment and control sectors that predated the PLI scheme, we would expect to find a significant effect even with a fake treatment date. The absence of such an effect strengthens the credibility of the main DiD result as reflecting the causal impact of the PLI scheme rather than a spurious correlation.



Additional robustness or pre-treatment trend graph.

8. Discussion

The point estimate of the DiD coefficient ($\delta = 0.88$) translates to an approximate 141% increase in exports in PLI-covered sectors relative to control sectors in the post-treatment period. Taken at face value, this is a large and economically significant effect. The raw data supports this in the case of Mobile Manufacturing: exports grew from approximately USD 3,837 million in 2019 (just before PLI) to USD 24,138 million in 2024 a roughly 6-fold increase. Medical Devices also showed consistent growth, from USD 316 million to USD 479 million over the same period.

However, the lack of statistical significance ($p = 0.42$) means we cannot confidently attribute this entire increase to the PLI scheme. The confidence interval around the estimate is wide, consistent with zero treatment effect at conventional levels. This should be interpreted in the context of the very small sample size rather than as strong evidence against a policy effect.

9. Limitations

Several limitations should be acknowledged:

1. Small sample: The panel has only 4 sectors and 8 years, yielding 32 observations. With 4 sector clusters, the clustered standard errors are unreliable. Wild cluster bootstrap methods (Cameron, Gelbach and Miller, 2008) or alternative inference approaches would be more appropriate but cannot meaningfully be applied with only 4 clusters.
2. Sector selection: The control group (Leather and Ceramics) differs substantially from the treatment group in absolute export scale. While the log transformation mitigates scale differences, there may be remaining structural differences that make these imperfect counterfactuals.
3. Confounding factors: The post-2020 period coincides not only with the PLI scheme but also with the COVID-19 pandemic, global supply chain restructuring (China+1 strategy), and India-specific industrial policy changes. Fully disentangling PLI effects from these concurrent factors is difficult with the available data.
4. Heterogeneous treatment effects: The TWFE estimator provides a weighted average treatment effect, which may mask important heterogeneity across sectors and over time. The event study results suggest the effect builds over time, which is consistent with the economic mechanism but also implies that a single aggregate DiD estimate may be insufficient.

10. Conclusion

This study evaluates the causal impact of India's Production Linked Incentive (PLI) scheme on export performance using a Difference-in-Differences methodology applied to a panel of four sectors spanning 2017-2024. The treatment sectors (Mobile Manufacturing and Medical Devices) are compared against control sectors (Leather Goods and Ceramics) not covered by the PLI scheme.

The main TWFE DiD estimate yields a coefficient of 0.88 (approximately 141% increase in exports), which is positive and economically significant in magnitude, though not statistically significant at conventional levels ($p = 0.42$) due to the small sample. The event study reveals no significant pre-treatment divergence between treated and control sectors in 2017 and 2018, providing empirical support for the parallel trends assumption. The post-treatment event study coefficients show a pattern of initial disruption (COVID-19 in 2020), followed by gradual recovery and a building positive trend from 2023-2024, consistent with the economic mechanism of the PLI scheme.

The placebo test, using a fake treatment date of 2018, returns an insignificant coefficient ($p = 0.36$), further validating the DiD design. Taken together, the results are suggestive of a positive PLI effect on export performance that accumulates over time, consistent with the intended policy design of incentivising incremental production and export growth over a five-year horizon.

Future research with a larger sector sample, monthly or quarterly data, and firm-level analysis would substantially improve the precision and reliability of these estimates.

References

- Angrist, J.D., and Pischke, J.S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Cameron, A.C., Gelbach, J.B., and Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414-427.
- Cameron, A.C., and Miller, D.L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317-372.
- Callaway, B., and Sant'Anna, P.H.C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200-230.
- Holland, P.W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*, 81(396), 945-960.
- Ministry of Commerce and Industry, Government of India (2020). *Production Linked Incentive Scheme Notification*. New Delhi.
- Rubin, D.B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688-701.
- Sun, L., and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175-199.